



Note:1 The exact OQL that can be used varies based on the tool and the JVM version. The best way to find out the OQL that can be used on a particular tool is by accessing the OQL of that particular tool. In the jhat tool, this can be obtained by accessing the following URL: <http://localhost:7000/oqlhelp/>

2) Based on the size of heap, OQL tools can quickly run out of memory while executing queries and searching the heap. It is recommended that the heap available to the tool itself be set to a large value. In jhat, the memory available can be increased using the `-J-mx1024m` option.

on why an object is still referenced, even though, as per the code in the program, the object can be garbage collected.

Shallow vs retained size

The memory profilers use the term shallow and retained sizes to indicate the memory occupied by objects. The retained size is the sum of shallow sizes of objects reachable from this object recursively, but excluding objects that are referenced by the GC Roots. For instance, in the following example:

```
public class A{
    private B b=new B();
}
public class B{
    private String s="Hello"
}
```

...the shallow size of an instance of A would not include the size of the instance of object B. It would just include the memory occupied by A. The retained size of A would include the size of B and also the size of any objects that B contains recursively.

Object Query Language (OQL)

Object Query Language (OQL) is a query language used by profiling tools to query memory. The original specification from Object Data Management Group (ODMG) was to query object databases. Most of the profiling tools support a version of OQL customised for the heap analysis. OQL resembles SQL, but the expressions in 'select' and 'where' clauses can be written in Javascript. It also has in-built

functions to find objects on heap, to find out the incoming and outgoing references of these objects. Some tools even provide advanced features, such as aggregation over a group of objects or the capability to find out the size of an object directly (`rsizeof`). The table on the previous page gives a few examples of commonly used OQL queries that are helpful in analysing objects on heap. 

References

- [1] Eclipse Memory Analysis Tool (MAT) www.eclipse.org/mat
- [2] Object Query Language - http://en.wikipedia.org/wiki/Object_Query_Language
- [3] IBM Heap Dumps http://publib.boulder.ibm.com/infocenter/eduaass/v1r1m0/index.jsp?topic=/com.ibm.jaas.was_v6/was/6.0.1/PD/WASv601_zOS_Heapdumps/player.html
- [4] HPROF: A Heap/CPU Profiling Tool - <http://docs.oracle.com/javase/7/docs/technotes/samples/hprof.html>
- [5] Path to GC Queries: http://pic.dhe.ibm.com/infocenter/isa/v4r1m0/index.jsp?topic=/2Fcom.ibm.java.diagnostics.memory.analyzer.doc%2Fpath_to_grooots.html
- [6] Java Tools: Jps: <http://docs.oracle.com/javase/7/docs/technotes/tools/share/jps.html>
Jstat: <http://docs.oracle.com/javase/7/docs/technotes/tools/share/jstat.html>
JMap: <http://docs.oracle.com/javase/7/docs/technotes/tools/share/jmap.html>
Jhat: <http://docs.oracle.com/javase/7/docs/technotes/tools/share/jhat.html>
- [7] Java Instrumentation package: <http://docs.oracle.com/javase/7/docs/api/index.html?java/lang/instrument/package-summary.html>
- [8] Visual GC help: <http://www.oracle.com/technetwork/java/visualgc-136680.html>
- [9] IBM Developer Works Community: <https://www.ibm.com/developerworks/community/>
- [10] Java HPROF: <http://docs.oracle.com/javase/7/docs/technotes/samples/hprof.html>
- [11] Portable Heap Dump: http://pic.dhe.ibm.com/infocenter/java7sdk/v7r0/index.jsp?topic=/2Fcom.ibm.java.win.70.doc%2Fdiag%2Ftools%2Fheapdump_phd_format.html

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